

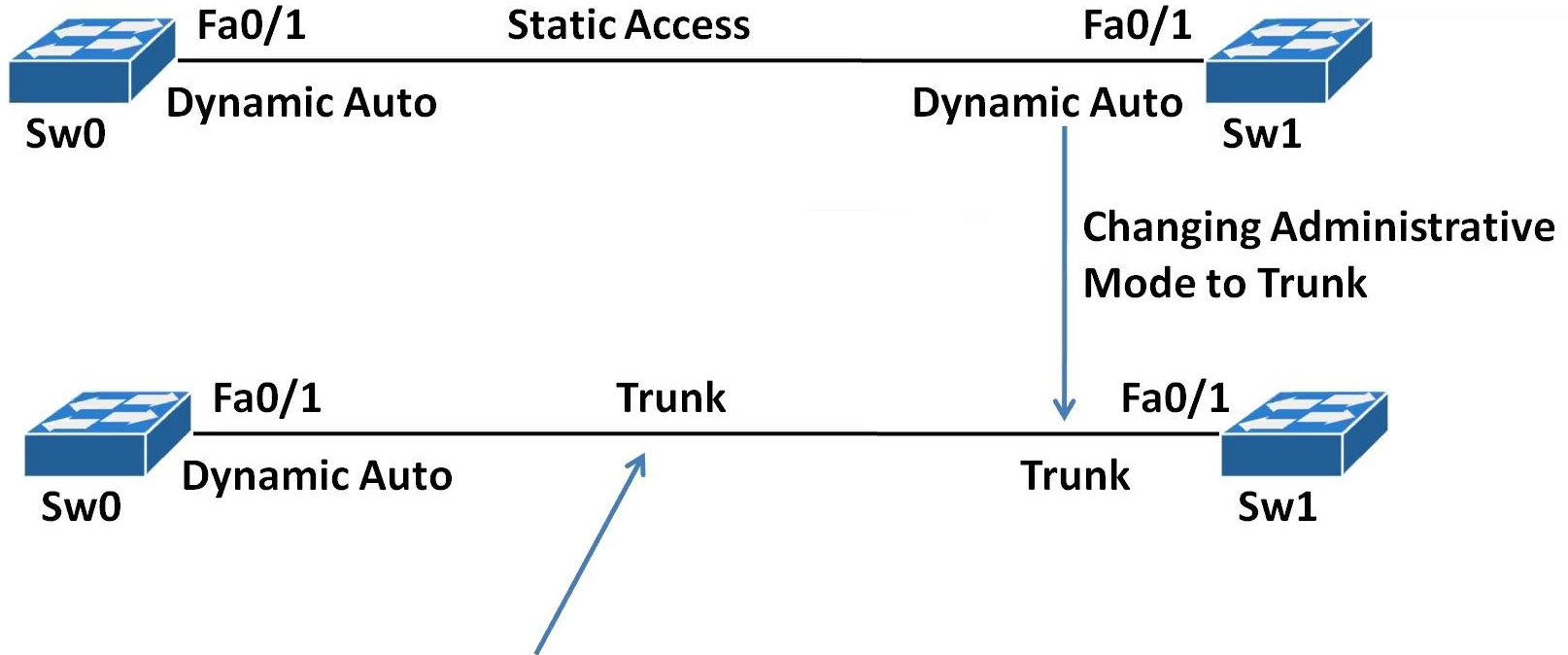
Dynamic Trunking Protocol:

DTP (Dynamic Trunking Protocol) is a Cisco proprietary networking protocol used to automate the configuration of trunk links between Cisco switches. Trunk links are connections between switches that carry traffic for multiple VLANs (Virtual Local Area Networks).

Purpose: DTP simplifies the process of setting up trunk links by dynamically negotiating trunking on a link between two switches and choosing the appropriate encapsulation method (such as IEEE 802.1Q or ISL).

Explanation of DTP

By Default All the interfaces of switches are in Dynamic Auto mode



Operational mode(Link) Automatically Changed to the Trunk mode

Modes of DTP

Administrative Mode (Interface Mode):

Access: Forces the port to be a non-trunking, single VLAN interface.

Trunk: Forces the port to be a trunk link.

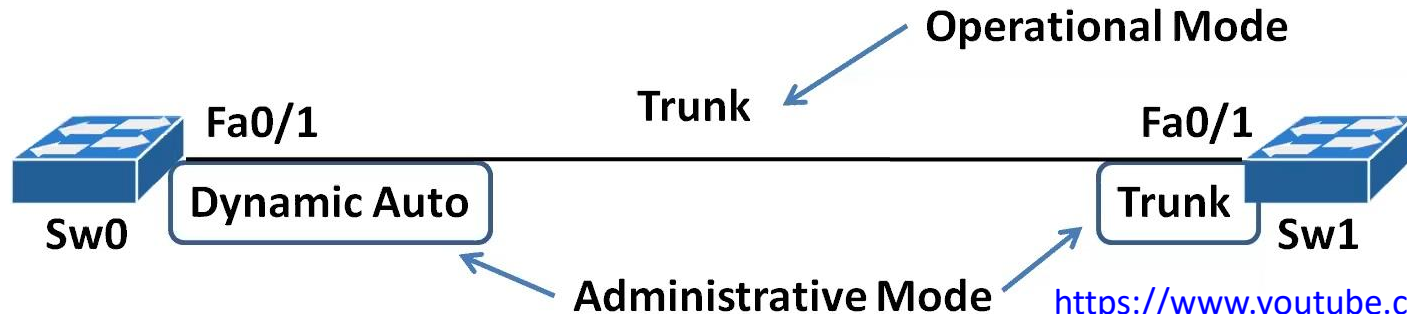
Dynamic Desirable: Actively attempts to convert the link to a trunk link.

Dynamic Auto: Passively waits to be converted to a trunk link

Operational Mode (Mode of Link)

Trunk Mode: Multiple VLAN Information Can be forwarded

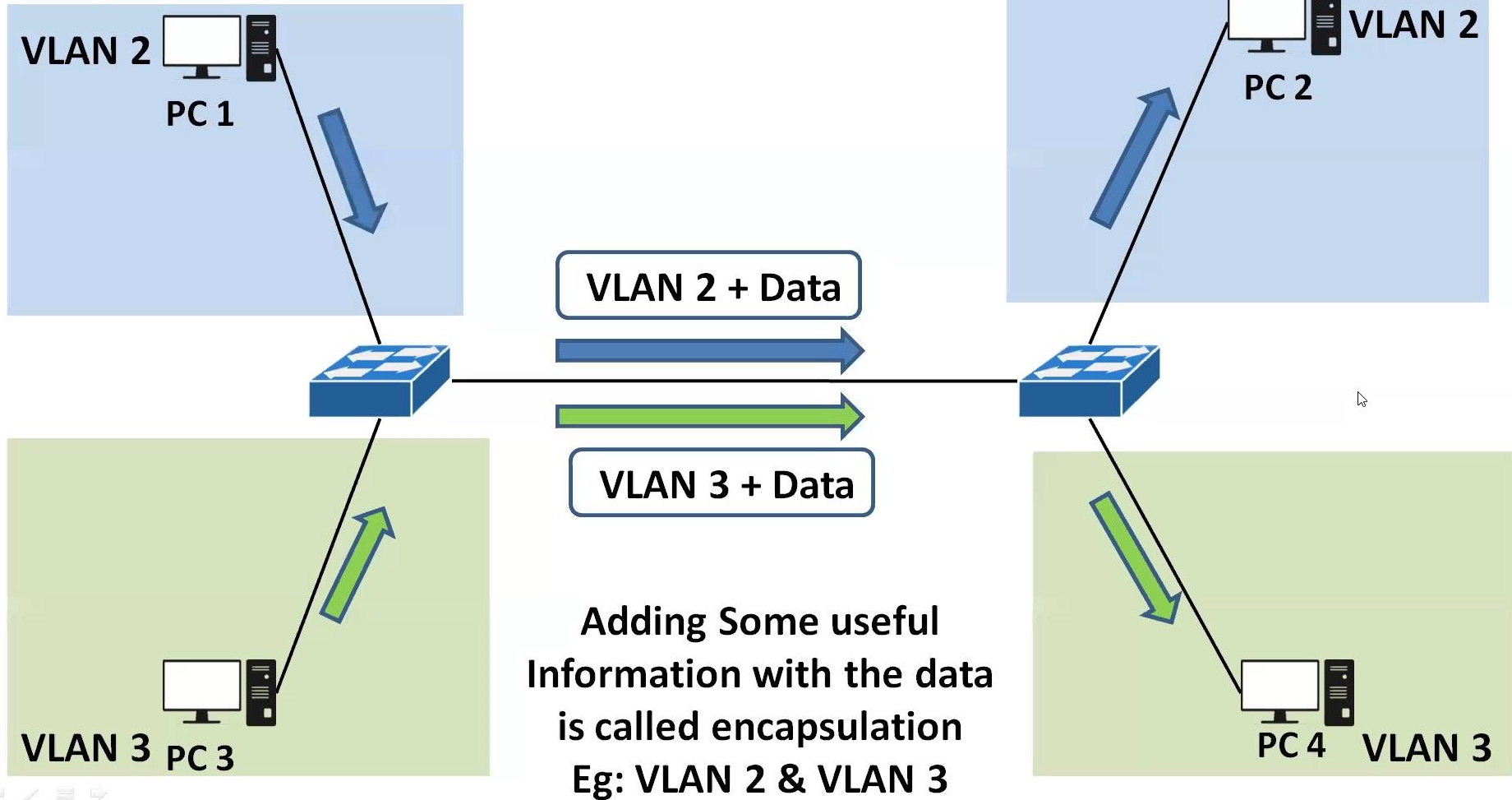
Static Access Mode: Multiple VLAN Information Cannot be forwarded



Possible Combination of DTP:

Administrative Mode	Administrative Mode	Operational Mode
Access	Access	Static Access
Access	Trunk	Static Access
Access	Dynamic Auto	Static Access
Access	Dynamic Desirable	Static Access
Dynamic Auto	Dynamic Auto	Static Access
Trunk	Trunk	Trunk
Trunk	Dynamic Auto	Trunk
Trunk	Dynamic Desirable	Trunk
Dynamic Auto	Dynamic Desirable	Trunk
Dynamic Desirable	Dynamic Desirable	Trunk

Encapsulation in DTP:



Types of Encapsulation

IEEE 802.1Q (Dot1Q)

- It inserts a 4-byte VLAN tag into the original Ethernet frame, specifying the VLAN to which the frame belongs.
- It is widely used and supported by most networking vendors, making it a preferred choice for mixed-vendor environments.

2. ISL (Inter-Switch Link)

- ISL encapsulates the original Ethernet frame by adding a 26-byte header and a 4-byte CRC trailer, creating a larger frame. This header contains VLAN information.
- Limited to Cisco equipment, which can be a constraint in heterogeneous networks.

Configuration:

Switch(config)# interface GigabitEthernet0/1

Switch(config-if)# switchport trunk encapsulation dot1q

Switch(config-if)# switchport mode trunk

